Stability of Periodic Orbits in the Elliptic, Restricted Three-Body Problem

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A systematic study has been made of periodic orbits in the two-dimensional, elliptic, restricted three-body problem. All ranges of eccentricities, from 0 to 1, and mass-ratios, from 0 to ½, have been investigated. Eleven hundred periodic orbits have been obtained. It is concluded that the elliptic problem behaves in a way which is completely different from the circular problem. The main difference is in the stability properties of the periodic orbits. Because of the nonexistence of the Jacobi integral (the elliptic problem is not conservative), the characteristic equation of the monodromy matrix does not have a pair of unit roots, in general. The stability is defined by two real numbers (stability indices) rather than one. Because of that, there are seven general classes of periodic orbits, according to their stability properties. The stability of the periodic orbits has been determined by numerically integrating the variational equations with a recurrent power series method. The results are in contrast with the circular problem, where there are only three classes of orbits (stability, even instability, and odd instability): in the elliptic problem there are one stable class and six unstable classes. The elliptic, restricted three-body problem can be considered as the prototype of all nonintegrable, nonconservative Hamiltonian systems, and in this paper, probably for the first time, a classification of the multipliers is given for these systems.

I. Introduction

THIS article will consider the stability of periodic orbits in the elliptic, rather than circular, restricted problem of three bodies. The problem of finding periodic orbits in the circular, restricted, three-body problem has been very extensively studied in celestial mechanics. It is well known that in the circular problem, given a fixed mass-ratio, there exist families of periodic orbits continuously parametrized by the Jacobi constant, for which the period varies in a continuous way. The stability of these periodic orbits and all the related concepts such as the variational equations, the eigenvalues of the fundamental matrix, and the characteristic exponents have been very extensively studied.

However, the configurations found in the solar system are near to a three-body situation with orbits with nonzero eccentricities, and the elliptic problem is a better model than the circular one. For instance, for the motion of a satellite in the earth-moon system, as a first approximation, we may assume that the moon moves around the earth in circular motion, but as a much better approximation, we should include the eccentricity of the moon's orbit in the problem.

On the other hand, the elliptic, restricted, three-body problem is characterized by some fundamental differences from the circular problem, in such a way that the solutions appear to behave in a completely different way. Some of the most fundamental features of the elliptic restricted problem are that there is no Jacobi integral, that the time is explicitly present in the equations of motion even when a suitable rotating coordinate system is used, that there is no pair of zero characteristic exponents, and that there are only isolated periodic orbits, with a well-determined period, rather than families with continuously varying periods. The consequence of these facts is that we are in presence of a dynamical problem which is very different from the circular problem, and it is our opinion that this situation has not yet been studied enough.

For the foregoing reasons, we have started a systematic study of orbits in the elliptic restricted problem. We have supported this study by extensive numerical experiments, using classical numerical integration on the computer. We are describing here some of our results, mainly those results which are more directly related to the linear stability of the periodic orbits and the properties of the characteristic exponents of these orbits.

II. Equations of Motion for the Elliptic Problem

Let us first refer the problem to an inertial barycentric system of axes $(\bar{x}, \bar{y})$. The two heavy masses or primaries $m_1 = 1 - \mu$ and $m_2 = \mu$ revolve on elliptic orbits around their barycenter in such a way that at any instant t their coordinates are

$$\begin{aligned}
\bar{x}_1 &= -\mu(\cos E - e) = -\mu r \cos v \\
\bar{y}_1 &= -\mu(1 - e^2)^{1/2} \sin E = -\mu r \sin v \\
\bar{x}_2 &= (1 - \mu)(\cos E - e) = (1 - \mu) r \cos v \\
\bar{y}_2 &= (1 - \mu)(1 - e^2)^{1/2} \sin E = (1 - \mu) r \sin v
\end{aligned} \tag{1}$$

The angles E and v are, respectively, the eccentric and true anomalies, and r is the distance between the two primaries:

$$r = 1 - e \cos E = (1 - e^2)/(1 + e \cos v) \tag{2}$$

We have thus used here a system of canonical units of length and time which are such that the semimajor axis a and mean motion n of the primaries are unity. We also can see that the origin of the time is chosen in such a way that, at $t = 0$, the two primaries are on the x axis and are at their minimum distance r (periapsis).

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The equations of motion for the satellite are then

$$\bar{x}'' = -(1-\mu)(\bar{x}-\bar{x}_1)/r_1^3 - \mu(\bar{x}-\bar{x}_2)/r_2^3$$
$$\bar{y}'' = -(1-\mu)(\bar{y}-\bar{y}_1)/r_1^3 - \mu(\bar{y}-\bar{y}_2)/r_2^3 \tag{3†}$$

The quantities r_1 and r_2 are the distances from the satellite to the primaries;

$$r_1^2 = (\bar{x}-\bar{x}_1)^2 + (\bar{y}-\bar{y}_1)^2$$
$$r_2^2 = (\bar{x}-\bar{x}_2)^2 + (\bar{y}-\bar{y}_2)^2 \tag{4}$$

The equations of motion (3) derive from the following Lagrangian function:

$$L = \tfrac{1}{2}(\bar{x}'^2 + \bar{y}'^2) + (1-\mu)/r_1 + \mu/r_2 \tag{5}$$

We introduce this Lagrangian here because we will use it in the following development for several coordinate transformations. It is indeed possible to make some changes of coordinates in such a way that, relative to the new axes, the primaries take on fixed positions. We have found this system of axes very convenient for the study of the elliptic problem.

We will now introduce a system of coordinates (ξ,η) which is barycentric and which is rotating with a variable angular velocity, in such a way that the rotation angle is at any instant equal to the true anomaly v;

$$x = \xi \cos v - \eta \sin v, \qquad y = \xi \sin v + \eta \cos v \tag{6}$$

In this system of axes the primaries are permanently on the ξ axis, at the coordinates

$$\xi_1 = -\mu r, \quad \xi_2 = (1-\mu)r, \quad \eta_1 = 0, \quad \eta_2 = 0 \tag{7}$$

They are thus oscillating on the ξ axis.

The Lagrangian for the motion of the satellite now becomes

$$L = \tfrac{1}{2}(\xi'^2 + \eta'^2) + \tfrac{1}{2}(\xi^2+\eta^2)v'^2 + (\xi\eta' - \xi\eta')v' + (1-\mu)/r_1 + \mu/r_2 \tag{8}$$

where the derivative v' of the true anomaly is known from the two-body problem to be

$$v' = (1-e^2)^{1/2}/r^2 = p^{1/2}/r^2 \tag{9}$$

Next we introduce a system of rotating pulsating coordinates (x,y) derived from the rotating coordinates (ξ,η) by

$$\xi = rx, \qquad \eta = ry \tag{10}$$

The primaries will then have constant coordinates

$$x_1 = -\mu, \quad x_2 = (1-\mu), \quad y_1 = 0, \quad y_2 = 0 \tag{11}$$

The Lagrangian of the problem becomes

$$L = \frac{r^2}{2}(x'^2 + y'^2) + rr'(xx' + yy') + (x^2+y^2) + \frac{r'^2}{2}(x^2+y^2) + \frac{p}{2r^2}(x^2+y^2) + p^{1/2}(xy' - yx') + \frac{1}{r}\left(\frac{1-\mu}{r_1} + \frac{\mu}{r_2}\right) \tag{12}$$

It is possible to simplify considerably this Lagrangian by making a change of independent variable in it, from time t to true anomaly v. A convenient technique for making a change of independent variable in a Lagrangian is described in the author's 1967 article.[1] In the Lagrangian (12), if we make a change of independent variable according to

$$dt = 1/p^{1/2}r^2 dv \tag{13}$$

which is equivalent to (9), we have to replace the terms in x' by $p^{1/2}\dot{x}/r^2$, and then multiply the Lagrangian by $r^2/p^{1/2}$. After simplifying some terms, we find that, except for a factor $p^{1/2}$, the Lagrangian (12) transforms into

$$L = \frac{1}{2}(\dot{x}^2 + \dot{y}^2) + (x\dot{y} - y\dot{x}) + \frac{r}{2p}(x^2+y^2) + \frac{r}{p}\left(\frac{1-\mu}{r_1^2} + \frac{\mu}{r_2^2}\right) \tag{14}$$

The corresponding equations of motion in this system are

$$\ddot{x} - 2\dot{y} = \frac{r}{p}\left(x - (1-\mu)\frac{x-x_1}{r_1^3} - \mu\frac{x-x_2}{r_2^3}\right)$$
$$\ddot{y} + 2\dot{x} = \frac{r}{p}\left(y - (1-\mu)\frac{y}{r_1^3} - \mu\frac{y}{r_2^3}\right) \tag{15}$$

together with a differential equation for t

$$\dot{t} = r^2/p^{1/2} \tag{16}$$

The function r which is present in (15) has to be considered as a function of the new independent variable v. For the purpose of numerical integration with recurrent power series, we have found it convenient to define r by its differential equation (in v)

$$\ddot{r} = 2\frac{\dot{r}^2}{r} + r\left(1 - \frac{r}{p}\right) = \frac{-2}{p}r^3 + \frac{3}{p}r^2 - r \tag{17}$$

rather than with the use of (2). Considering (15–17) together, we now have a seventh-order system of differential equations which is autonomous (i.e., which does not contain the independent variable v explicitly). We have used this system for programming and integration on a digital computer. A more detailed description of the basic equations of motion and the principal properties of the elliptic three-body problem is given in V. Szebehely's *Theory of Orbits*.[3]

As a numerical integration technique we have used the so-called Steffensen method with recurrent power series. To set up the necessary programs, we have closely followed the ideas of a publication by A. Deprit on the circular three-body problem,[2] both for the equations of motion themselves and for the corresponding variational equations. To make the system (15–17) suitable for the recurrent power series approach, several transformations have been made. Some redundant variables are introduced in such a way that we finally have 15 dependent variables for the equations. (We have actually programed the equations of motion for the three-dimensional program, and we then have 18 variables.) The new variables (or parameters) of the problem are called $P_1, P_2, \ldots, P_{18}$.

III. Variational Equations of the Elliptic Restricted, Three-Body Problem

The variational equations play an important role because, through their solution, we can get the partial derivatives of the variables $(xy\dot{x}\dot{y})$ with respect to their initial conditions $(x_0y_0\dot{x}_0\dot{y}_0)$. The 16 partial derivatives are obtained by numerically integrating four times the variational equations. These four independent solutions of the variational equations can be considered as four columns of a 4×4 matrix, called the fundamental matrix. In our computations the initial conditions form a unit matrix, and we call the particular fundamental matrix which is then obtained the principal fundamental matrix, but it is also known as the matrizent or resolvent.

The principal fundamental matrix, containing the 16 partial derivatives will be needed for two important reasons: it is needed for the differential correction process to find the

† Here we use primes to indicate the time derivatives, but later we will use dots to indicate derivatives with respect to the true anomaly v.

periodic orbits and it also is needed to determine the first-order stability of the orbit by consideration of the eigenvalues of the matrix. If the equations of motion are written in the form

$$\frac{dx}{dv} = \dot{x}, \quad \frac{dy}{dv} = \dot{y}, \quad \frac{d\dot{x}}{dv} = f(x,y,\dot{y}), \quad \frac{d\dot{y}}{dt} = g(x,y,\dot{x}) \tag{18}$$

then we can write the variational equations as

$$\left.\begin{aligned} d\delta v/dv &= \delta\dot{x}, \qquad d\delta v/dv = \delta\dot{y} \\ \frac{d\delta\dot{x}}{dv} &= \frac{\partial f}{\partial x}\,\delta x + \frac{\partial f}{\partial y}\,\delta y + \frac{\partial f}{\partial \dot{y}}\,\delta\dot{y} \\ \frac{d\delta\dot{y}}{dv} &= \frac{\partial g}{\partial x}\,\delta x + \frac{\partial g}{\partial y}\,\delta y + \frac{\partial g}{\partial \dot{x}}\,\partial\dot{x} \end{aligned}\right\} \tag{19}$$

The differential partial derivatives which are present in (19) are

$$\left.\begin{aligned} \partial f/\partial\dot{y} &= -\partial g/\partial\dot{x} = +2 \\ \frac{\partial f}{\partial x} &= \frac{r}{p}\left[1 - V_3 + 3\left(m_1\,\frac{(x-x_1)^2}{r_1^5} + m_2\,\frac{(x-x_2)^2}{r_2^5}\right)\right] \\ \frac{\partial f}{\partial y} &= \frac{\partial g}{\partial x} = \frac{3ry}{p}\left[m_1\,\frac{(x-x_1)}{r_1^5} + m_2\,\frac{(x-x_2)}{r_2^5}\right] \\ \frac{\partial g}{\partial y} &= \frac{r}{p}\left[1 - V_3 + 3y^2\left(\frac{m_1}{r_1^5} + \frac{m_2}{r_2^5}\right)\right] \end{aligned}\right\} \tag{20}$$

with

$$V_3 = m_1/r_1^3 + m_2/r_2^3 \tag{21}$$

In order to obtain simultaneously with the solution of the equations of motion four independent solutions of the variational equations, we have defined, besides the 18 parameters related to the equations of motion, 32 new parameters P_{19}–P_{50}. The 16 variables P_{19}–P_{34} give the four solutions of the variational equations, so that the principal fundamental matrix R is formed in the following way with these variables:

$$R = \begin{vmatrix} P_{19} & P_{23} & P_{27} & P_{31} \\ P_{20} & P_{24} & P_{28} & P_{32} \\ P_{21} & P_{25} & P_{28} & P_{33} \\ P_{22} & P_{26} & P_{30} & P_{34} \end{vmatrix} \tag{22}$$

The other 16 variables P_{35}–P_{50} are auxiliary quantities which are used in the recurrent power series method in order to simplify the recurrence relations that are obtained for the coefficients of the series. Using a large number of auxiliary variables, we can set up the chain of computations in such a way that no more than two series ever have to be multiplied together, and very simple recurrence relations for the coefficients are obtained. The formulation that we have used is certainly not unique, and, in fact, a formulation with fewer variables is possible. On the other hand, the redundancy of the variables can be used for the purpose of checking the precision of the integration. The recurrence relations that are obtained are very similar to those which are in Ref. 2 for the circular problem, for instance, and for this reason we have not found it useful to reproduce the rather lengthy but elementary formulas here.

IV. Some Important Differences between the Elliptic and the Circular Restricted Three-Body Problems

We will briefly describe here some important features of the elliptic problem. The first is that the elliptic problem has no Jacobi integral (or energy integral) as the circular problem does. The absence of a first integral in the elliptic problem has far-reaching consequences, the most important of them being that there will be no more pair of unit eigenvalues for the fundamental matrix R. In the circular case it is known that the four eigenvalues of R are of the form

$$\lambda,\ 1/\lambda,\ +1,\ +1 \tag{a}$$

However, because of the absence of the energy integral, in the elliptic case the eigenvalues of the fundamental matrix R will be of the form

$$\lambda,\ 1/\lambda,\ \mu,\ 1/\mu \tag{b}$$‡

More detailed comments on this fact will be given later in this article.

A second important characteristic of the elliptic problem is that this problem has no natural families of periodic orbits, in the sense of the circular problem. The equations of motion, in the rotating as well as in the inertial coordinate systems, contain the time explicitly, for instance through the true or eccentric anomalies v or E. In both cases the equations of motion contain $\cos v$, $\sin v$ or $\cos E$, $\sin E$. Because this problem is what some authors call nonautonomous and others nonconservative, any periodic solution must have a period which is an integer multiple of the period of the periodic functions $\cos v$ and $\sin v$. In other words, the period of any periodic orbit of the elliptic, restricted problem must be 2π or a multiple of it. The period of rotation of the rotating axes being also 2π, a periodic orbit of the elliptic problem will be periodic with respect to the inertial axes as well as the rotating axes. Except for some details, related, for instance, to the convenience of numerical integration, it is thus not essential to use rotating axes in this problem. Periodic orbits could be studied just as well with respect to the inertial axes.

The question of symmetric orbits (with respect to the *ox* axis, or more precisely the *syzygy* axis), which is so important in the circular problem, still has all its value in the elliptic problem, except that the periodicity criterion will be more strict here. In the circular problem we can state what we call the "weak" periodicity criterion. "An orbit which has two perpendicular crossings with the *syzygy* axis is periodic."[4] The times of crossings are unimportant in the circular problem, and families of periodic orbits with a continuously varying period are known to exist. In the elliptic problem, the criterion for symmetric periodic orbits, which we call the "strong" periodicity criterion, may be stated as follows:

> For an orbit to be periodic it is sufficient that it has two perpendicular crossings with the syzygy-axis, and that the crossings happen at moments when the two primaries are at an apse, (i.e., at maximum or minimum elongation, or apoapsis and periapsis).[4]

The time between two apses in the two-body problem being an integer multiple of π, say $k\pi$, the period will then be $2k\pi$. Of course, the preceding criterion for periodicity is a sufficient but not necessary condition, as nonsymmetric orbits may also exist.

In our research on the elliptic problem we have investigated some cases corresponding to the integers $k = 1, 2, 3, 4, 5$. The aforementioned strong periodicity criterion was essentially given by Moulton in his book on periodic orbits.[4]

We have used the following procedure to find periodic orbits in the elliptic problem. We have taken the circular problem as a first approximation. In this problem we had already classified in families some 4000 periodic orbits, mostly corresponding to the earth-moon mass-ratio.[5] Out of this large collection of periodic orbits in the circular problem, the isolated periodic orbits with period $2k\pi$ ($k = 1, 2, 3, 4, 5$) have then been extracted by interpolation, in an automatic way with the use of a computer program. In this way some 150 periodic orbits, with the right period, which we call syn-

‡ The symbol μ which is used here should not be confused with the mass-ratio, also represented by μ.

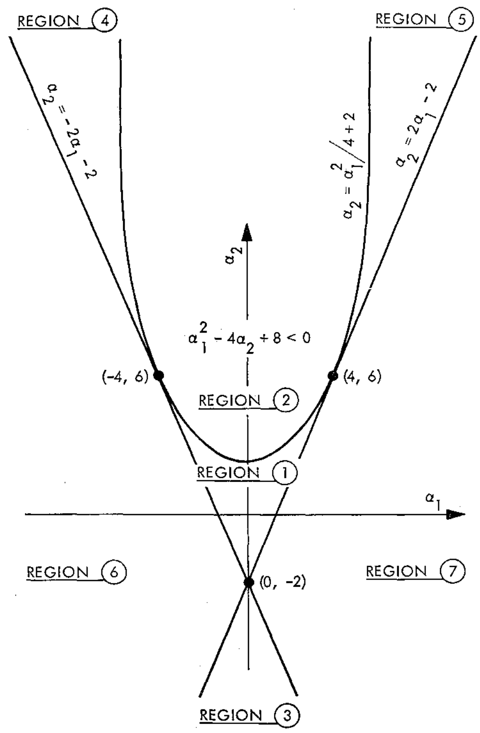

Fig. 1 Seven stability regions.

chronous or commensurable periodic orbits, have been obtained. Some of these orbits have then been used as starters for a numerical continuation parametrized by the eccentricity e. We have seen that, for some orbits of the circular problem, a prolongation to the elliptic problem is possible for all eccentricities up to and including $+1$ (even for all mass-ratios μ, from 0 to $\frac{1}{2}$). For all those periodic orbits of the elliptic problem, the period is kept constant, and the planetoid is synchronized with the two primaries.

To find the symmetric periodic orbits each time e is slightly incremented, linear differential corrections in two variables can be used. Let us suppose that an orbit with initial conditions $(x_0,0,0\dot{y}_0)$ has been integrated over an interval 0 to $k\pi$ for v and has given, at the expected crossing, the values $(x_F,y_F,\dot{x}_F,\dot{y}_F)$ for the four variables. Corrections to the initial conditions, Δx_0, $\Delta \dot{y}_0$, are then obtained by solving the two following linear equations:

$$\frac{\partial y}{\partial x_0}\Delta x_0 + \frac{\partial y}{\partial \dot{y}_0}\Delta \dot{y}_0 = -y_F, \quad \frac{\partial \dot{x}}{\partial x_0}\Delta x_0 + \frac{\partial \dot{x}}{\partial \dot{y}_0}\Delta \dot{y}_0 = -\dot{x}_F \quad (23)$$

This process will generally converge in less than about five iterations, under the condition that the expected periodic orbit exists and that a sufficiently good initial guess is available. The four necessary partial derivatives in (23) are given by the numerical integration of the variational equations.

In some cases the determinant of (23) may be small, and then the convergence of the iterations may become poor. In most cases these difficulties have been resolved by the use of a sophisticated matrix inversion program prepared at the Jet Propulsion Laboratory by C. Lawson. This program uses a least-squares approach in cases where the determinant of the matrix is too small.

A third important difference between the elliptic and the circular problem is that the elliptic problem contains two parameters, μ and e, the mass-ratio of the primaries and the eccentricity of their orbits. The circular problem corresponds to $e = 0$ and has thus only one parameter, μ. As we have said previously, when e and μ are fixed, then only isolated periodic orbits exist in the elliptic problem. However, when we allow e and μ to vary, then a periodic orbit eventually can be prolongated in a double infinity of periodic orbits. We have found that some periodic orbits do exist for all values of e and μ (from 0 to 1). The periodic orbits that are described below all correspond to a fixed value of μ and are thus one-parameter families, e being the parameter; however, the computer programs that have been prepared are capable of perturbing both e or μ in a continuous way and differentially correcting to find the new periodic orbits.

Most of our integrations have been done with constant μ and increasing e, starting from $e = 0$. However, it can be seen that, for every symmetric periodic orbit with $e = 0$, there are two ways of prolongating it to the elliptic problem. The first way considers that at $t = 0$ we have $v = 0$, i.e., we have at $t = 0$ a periapsis for the two primaries. In the second possibility we have $v = \pi$ when $t = 0$, and we suppose thus that the two primaries are at apoapsis at the moment of the first crossing at $t = 0$. When $e = 0$, the forms of the orbits are identical, but there is a phase delay of a half-revolution between the two primaries and the satellite. When the eccentricity e is increased to nonzero values, the forms of the orbits are different and two different families are thus generated. These two families join at the point $e = 0$. This will be clearly seen in what follows, in relation with the stability study. We have decided to call periapsis orbits those periodic orbits which start with $v = 0$, and apoapsis orbits those which start when $v = \pi$.

V. Stability of Periodic Orbits in the Elliptic, Restricted Three-Body Problem

The stability of the periodic orbits depends on the behavior of the eigenvalues of the principal fundamental matrix R. We will consider here only the first-order or linear stability. Let us designate by

$$\lambda_1,\lambda_2,\lambda_3,\lambda_4 \quad \text{(c)}$$

the four eigenvalues of R, corresponding to a periodic orbit, and evaluated at the end of one complete revolution, i.e., after an interval $2k\pi$ of the true anomaly v. To these eigenvalues λ_i correspond the characteristic exponents α_i defined by (T being the period $2k\pi$)

$$\lambda_i = e^{\alpha_i T} \quad (24)$$

up to the addition of an arbitrary multiple of $2\pi i$. We will say that the orbit is stable in the first order if and only if the real parts of all the α_i are less than or equal to zero.

However, the eigenvalues of the fundamental matrix R have some remarkable properties which will allow us to state the stability criterion in a different form. The most interesting property of the eigenvalues of the matrix R is that they occur in reciprocal pairs. We will first prove this theorem.

The variational equations (19) may be written in a matrix form

$$dX/dv = AX \quad (25)$$

where the 4×4 matrix can be written as

$$A = \begin{vmatrix} 0 & I \\ a & 2J \end{vmatrix} \quad (26)$$

The matrix I is the 2×2-unit matrix; a is a symmetric (nonconstant) 2×2 matrix with the partial derivatives occurring in the variational equations (19), and J is the 2×2 matrix

$$J = \begin{vmatrix} 0 & +1 \\ -1 & 0 \end{vmatrix} \quad (27)$$

If we now define the 4×4 matrix S by

$$S = \begin{vmatrix} 0 & +1 \\ -I & 2J \end{vmatrix} \quad (28)$$

then we can verify directly that the matrix A satisfies the relation

$$SA^TS^{-1} = -A \tag{29}$$

This relation will allow us to obtain a simple expression for the inverse of the fundamental matrix R^{-1}. We know that the matrix R satisfies the differential equation (25)

$$dR/dv = AR \tag{30}$$

and also that the inverse of R satisfies the adjoint equation

$$dR^{-1}/dv = -R^{-1}A \tag{31}$$

Using (29) we now can see directly that the adjoint equation (31) is verified if R^{-1} has the following expression:

$$R^{-1} = SR^TS^{-1} \tag{32}$$

The superscript T here indicates the transpose operation. The expression (32) for the inverse of R shows that R^{-1} and R^T, or R, have the same eigenvalues. Thus when an eigenvalue of R is λ, then $1/\lambda$ is also an eigenvalue. Zero eigenvalues are excluded here, of course, because the determinant of both matrices R and R^{-1} is $+1$. As a consequence we may thus write the four eigenvalues of R in the form

$$\lambda, 1/\lambda, \mu, 1/\mu \tag{d}$$

This property is true for all values of the independent variable, and for all solutions, periodic and nonperiodic, as it has been shown in more detail in Ref. 6. We will use this only for periodic solutions and only for the value $2k\pi$ of the independent variable v. The characteristic exponents related to the eigenvalues (d) may be designated by $(\alpha, -\alpha, \beta, -\beta)$;

$$\lambda = e^{\alpha T}, \qquad \mu = e^{\beta T} \tag{33}$$

We may now state the stability criterion in a different way: we have stability if and only if α and β are pure imaginary, i.e., if λ and μ are both on the unit circle. We will thus have to discuss the location of the four eigenvalues of R with respect to the unit circle.

As a consequence of the form (d) of the roots, the characteristic equation of R may be written in the following form:

$$s^4 + a_1s^3 + a_2s^2 + a_1s + 1 = 0 \tag{34}$$

The two coefficients of s^4 and s^0 are thus equal as well as the two coefficients of s^3 and s. We have used these two properties very extensively for checking the precision of the numerical computations. In most cases we have obtained for these two relations a precision of at least 10^{-10}. As an important consequence of (34), we see that the configuration of roots, and thus the stability, will depend exclusively on the two coefficients a_1 and a_2. In other words, these two invariant coefficients a_1 and a_2 are associated to a periodic orbit, and each periodic orbit may be represented by a point in the (a_1, a_2) plane. We will call a_1 and a_2 the stability coefficients of the orbit. Also, we will call stability indices of the orbit the two numbers

$$k_1 = \lambda + 1/\lambda, \qquad k_2 = \mu + 1/\mu \tag{35}$$

This is in contrast with the circular problem, where the stability is determined by one single stability index k, rather than two, because of the presence of two unit-roots in the circular problem. In the elliptic problem the stability will be discussed in the (a_1, a_2) plane. The stability coefficients a_1 and a_2 are always real, of course, as they are obtained by taking subdeterminants of the real matrix R. In the circular problem, $\mu = 1/\mu = 1$ and $k_2 = 2$.

The characteristic equation (34) may be written in the form

$$(s - \lambda)(s - 1/\lambda)(\beta - \mu)(s - 1/\mu) = 0 \tag{36}$$

and this gives us the relation between the stability coefficients (a_1, a_2) and the stability indices (k_1, k_2);

$$\begin{aligned} a_1 &= -(\lambda + 1/\lambda + \mu + 1/\mu) = -(k_1 + k_2) \\ a_2 &= 2 + (\lambda + 1/\lambda)(\mu + 1/\mu) = 2 + k_1k_2 \end{aligned} \tag{37}$$

This shows us that k_1 and k_2 are the roots of a second-degree equation

$$X^2 + a_1X + (a_2 - 2) = 0 \tag{38}$$

and we may write for k_1 and k_2 the expressions

$$\left.\begin{matrix} k_1 \\ k_2 \end{matrix}\right\} = \frac{a_1 \pm (a_1^2 - 4a_2 + 8)^{1/2}}{2} \tag{39}$$

When k_1 and k_2 have been found, then the roots λ and μ can be found again by solving second-degree equations: the solution of

$$x^2 - k_1x + 1 = 0 \tag{40}$$

will give λ and $1/\lambda$ by

$$\left.\begin{matrix} \lambda \\ 1/\lambda \end{matrix}\right\} = \frac{k_1 \pm (k_1^2 - 4)^{1/2}}{2} \tag{41}$$

and the solution of

$$y^2 - k_2y + 1 = 0 \tag{42}$$

gives μ and $1/\mu$;

$$\left.\begin{matrix} \mu \\ 1/\mu \end{matrix}\right\} = \frac{k_2 \pm (k_2^2 - 4)^{1/2}}{2} \tag{43}$$

The problem of finding the four roots of (34) is thus reduced to solving three second-degree equations. It is important, of course, to consider the discriminants of these three equations. For (39) we have a zero discriminant when

$$a_2 = a_1^2/4 + 2 \tag{44}$$

and this locus is a parabola in the (a_1,a_2) plane, for which (0,2) is the apex and the a_2 axis the symmetry axis. For the other two second-degree equations, we have zero discriminants when k_1 or k_2 take on the values ± 2. These conditions correspond in the (a_1,a_2) plane to the two straight lines

$$a_2 = 2a_1 - 2, \qquad a_2 = -2a_1 - 2 \tag{45}$$

These two lines happen to be tangents at the parabola (44), at the points ($a_1 = \pm 4$, $a_2 = 6$). Using these two lines and the parabola itself, the plane (a_1,a_2) is subdivided in seven distinct regions. We will now discuss the properties of the roots for all seven regions. We will see that one region gives stability and the six other regions instability. The seven regions of the (a_1,a_2) plane are shown in Fig. 1. We notice on this diagram that the circular problem, which has two unit roots and thus $k_1 = +2$ or $k_2 = +2$, corresponds here to the line $a_2 = -2a_1 - 2$. The configurations of roots corresponding to the seven regions are shown in Fig. 2. We have the following principal characteristics of the roots, according to their region in the (a_1,a_2) plane.

1) Region One: $a_1^2 - 4a_2 + 8$ is positive. k_1 and k_2 are reals and $k_1^2 < 4$, $k_2^2 < 4$. λ and $1/\lambda$ are complex conjugates with $\lambda^2 + 1/\lambda^2 = +1$. μ and $1/\mu$ are complex conjugates with $\mu^2 + 1/\mu^2 = +1$. All four multipliers are on the unit circle. We have *stability*.

2) Region Two: $a_1^2 - 4a_2 + 8$ is negative. k_1 and k_2 are complex conjugates. The four multipliers are complex and are not on the unit circle. The conjugate of λ is μ. The conjugate of $1/\lambda$ is $1/\mu$. We have instability, and we have called this type of instability *complex instability*.

3) Region Three: $a_1^2 - 4a_2 + 8$ is positive. k_1 and k_2 are reals and $k_1^2 > 4$, $k_2^2 > 4$. λ and $1/\lambda$ are reals. μ and $1/\mu$ are reals and have signs which are opposite to the signs of λ and $1/\lambda$. Thus there are four real multipliers, two being

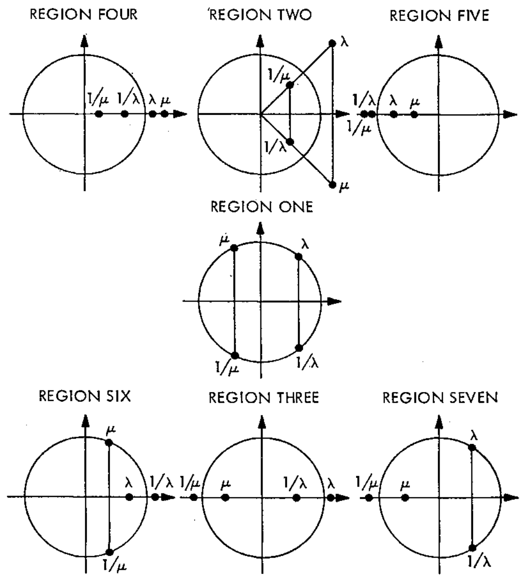

Fig. 2 Configuration of roots for each stability region.

positive and two negative. We have instability; we call this type *even-odd instability*.

4) Region Four: $a_1^2 - 4a_2 + 8$ is positive. k_1 and k_2 are reals and $k_1^2 > 4$, $k_2^2 > 4$; $k_1 > 0$, $k_2 > 0$. There are four real positive multipliers. We have instability: *even-even instability*.

5) Region Five: $a_1^2 - 4a_2 + 8$ is positive. k_1 and k_2 are reals and $k_1^2 > 4$, $k_2^2 > 4$; $k_1 < 0$, $k_2 < 0$. There are four real negative multipliers. We have instability: *odd-odd instability*.

6) Region Six: $a_1^2 - 4a_2 + 8$ is positive. k_1 and k_2 are real with $k_1^2 > 4$ and $k_2^2 > 4$. k_1 is real and positive and >2. λ and $1/\lambda$ are real and positive. μ and $1/\mu$ are complex conjugate, on the unit circle. We have instability: *even semi-instability*.

7) Region Seven: $a_1^2 - 4a_2 + 8$ is positive. k_1 and k_2 are real with $k_1^2 < 4$ and $k_2^2 > 4$. k_2 is real negative and <-2. μ and $1/\mu$ are real and negative. λ and $1/\lambda$ are complex conjugate, on the unit circle. We have instability: *odd semi-instability*.

VI. Numerical Examples

We have computed 15 families of periodic orbits. Most of these families have a fixed mass-ratio μ and variable eccentricity. We have computed the stability for six families only (about 620 orbits). These six families all have a fixed mass-ratio μ. They have been labeled as follows: 1) Family 7—periapsis ($\mu = 0.012155$, earth-moon mass-ratio); 2) Family 7—apoapsis ($\mu = 0.012155$, earth-moon mass-ratio); 3) Family 8—periapsis ($\mu = 0.5$, equal masses); 4) Family 8—apoapsis ($\mu - 0.5$, equal masses); 5) Family 11—periapsis ($\mu = 0.5$, equal masses); 6) Family 11—apoapsis ($\mu = 0.5$, equal masses). We will now give a short description for each one of the six families.

1) Family Seven—Periapsis

This family begins with one of the periodic orbits described in Ref. 5: an orbit of the family C of retrograde satellite orbits around the smaller primary m_2 in the circular restricted problem. The orbit which has been chosen is close to our orbit Nr 87 in Ref. 1, because this orbit has a period of 2π. The initial conditions, in rotating axes, may be given as follows: $x_0 = 0.15212027$, $\dot{y}_0 = 3.16076559$, $\mu = 0.012155$, $e = 0.0$. This family thus belongs to the earth-moon mass-ratio and has e as the variable parameter.

We have computed about 130 orbits in this family, with eccentricities e from 0 to 0.50. No orbits with higher eccentricity have been obtained yet because there is a collision with the larger primary just above this value of e. Although the orbits of this family belong to the class of "satellite orbits" in the circular problem, they are all quite large in shape and in fact they come closer to the larger primary m_1 than to m_2.

Good stability information has been obtained for the eccentricities up to 0.35 only. The orbits are all unstable and belong to region 6 in the stability diagram. However, they are all very close to the line $a_2 = -2a_1 - 2$ in this diagram. This is to be expected because we have essentially two-body orbits around the larger primary m_1 (in the inertial axes these orbits appear as ellipses), which are only weakly perturbed by the smaller primary. As a consequence, there still is an approximate integral of the motion, and two eigenvalues of the fundamental matrix are close to $+1$. Let us reiterate here that in the circular restricted three-body problem all the orbits belong to the straight line $a_2 = -2a_1 - 2$.

2) Family Seven—Apoapsis

This family begins with the same orbit as Family Seven—Periapsis. About 120 periodic orbits have been computed with eccentricity e from 0.0 to 0.99. This is thus a family which probably exists for all eccentricities. All the orbits have been computed in rotating (pulsating) axes using the Nechville transformation. The orbit corresponding to $e = 1.0$ would have to be computed in inertial axes.

Good stability information has been obtained only for the orbits with eccentricities below 0.75. All the orbits are stable and belong to region one. They are on a line which is again close to the line $a_2 = -2a_2 - 2$. When e increases, the orbits approach the separation point $(-4, 6)$ on the stability diagram.

3) Family Eight—Periapsis

This family begins at $e = 0.0$ with an orbit which has been taken from Stromgren's problem (circular, restricted three-body problem with equal masses). In fact we have taken the starting orbit from class g, in Bartlett's publications.[7] The period is 2π and the initial conditions for this orbit may be written as follows (in rotating axes): $x_0 = -0.4017933$, $\dot{y}_0 = +3.1437189$, $\mu = 0.5$, $e = 0.0$.

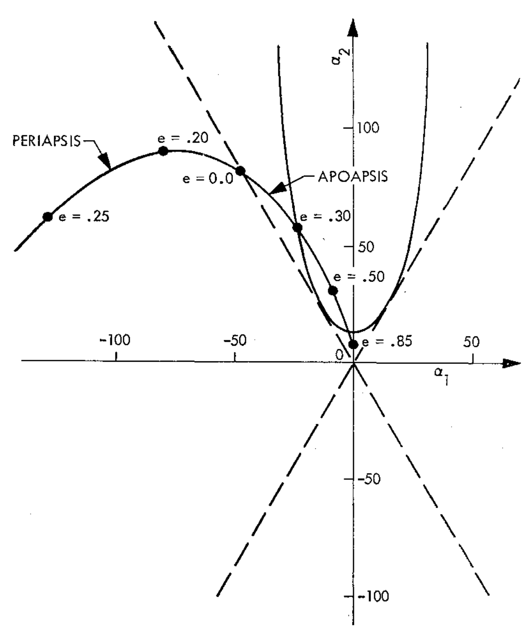

Fig. 3 Family 8 of periodic orbits.

About 120 orbits have been obtained in this family, with eccentricities ranging from 0.0 to 0.975. We do not know yet if the family can be prolongated all the way to $e = 1.0$. These last orbits with very high e would have to be computed in inertial axes; on the other hand, there seems to be an additional difficulty because the satellite has a close approach with one of the primaries when e approaches 0.97.

We have computed the stability only for the orbits with eccentricities up to 0.55. All these orbits are unstable and belong to Region Six in the stability diagram (even semi-instability).

4) Family Eight—Apoapsis

At $e = 0.0$, this family begins at the same orbit as Family Eight—Periapsis. About a hundred periodic orbits have been computed with eccentricities from 0 to 0.85. We have not obtained any orbits with eccentricities higher than 0.85 yet, because the convergence of the differential correction process was becoming increasingly slow. On the other hand, there seems to be a moderately close approach with one of the primaries for the last orbits which have been computed in this family.

In this family we find a remarkable evolution of the stability properties. We have been able to determine the roots of the fundamental matrix for all the orbits of this family, up to $e = 0.85$. The family starts in Region Four, passes in Region Two in the vicinity of $e = 0.30$, and stays in Region Two all the way up to $e = 0.85$, except that the last computed orbit $e = 0.85$ itself had just crossed the parabola $a_2 = a_1^2/4 + 2$, so that this orbit belongs to Region One and is stable. All the orbits of this family are thus unstable except the last one computed, which is stable. (Figure 3 depicts family 8 of periodic orbits.)

5) Family Eleven—Periapsis

This family also begins at $e = 0.0$, with an orbit with period 2π taken from Stromgren's problem, in fact, also from class g. The initial conditions in rotating axes are $x_0 = -0.07084826$, $\dot{y}_0 = +0.82832745$, $\mu = 0.5$, $e = 0.0$. We have computed 51 orbits in this family with e's from 0 to 0.453. We do not know at this moment whether or not this family goes much farther than $e = 0.543$. The last computed orbits show increasing difficulties in the convergence of the differential correction process. The last orbit, 0.453, was obtained with very slow convergence on the differential corrections (10 iterations), whereas for $e = 0.454$ there definitely was divergence. The loss of convergence is not due to the appearance of a close approach with one of the primaries.

The family begins with even instability of the circular problem at $e = 0$. The orbits then move into Region Four, even-even instability, with four real positive roots. The stability coefficient a_2 reaches a maximum of about 13,400 in the neighborhood of $e = 0.18$ and then decreases all the way to 1000 at $e = 0.45$. The stability coefficient a_1 increases continuously from -5000 to -200 when the eccentricity goes from 0.0 to 0.45. All the orbits of the family are thus unstable.

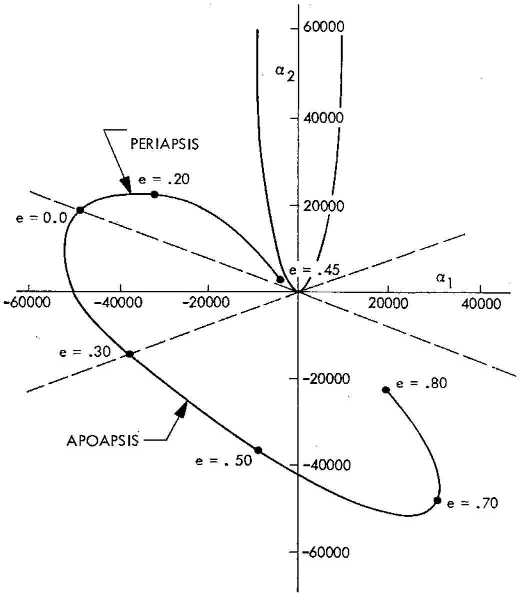

Fig. 4 Family 11 of periodic orbits.

6) Family Eleven—Apoapsis

The family begins with the same orbit as Family Eleven—Periapsis. About a hundred periodic orbits have been found, with $e = 0.0$ to $e = 0.895$. At this last value of e, there is a close approach with one of the primaries.

The stability of the orbits has been determined only up to $e = 0.80$. The stability coefficients a_1 and a_2 both go through a minimum: $a_1 = -5500$ at $e = 0.18$ and $a_2 = -5200$ at $e = 0.67$. From $e = 0.0$ to 0.30 all the orbits are in region six, and from $e = 0.30$ to $e = 0.80$ they are in region three. All the orbits of this family are thus unstable. (Figure 4 depicts family 11 of periodic orbits.)

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